University Physics A(2) 2014

Worksheet #9: Entropy (1)

Name (名字):   Student number (学号):

Problems Show all working.

(1) [M&I.12.P.39] The interatomic spring stiffness for copper (Cu) is known to be 28 N/m. The mass of one mole of copper is 0.064 kg.

In the following, you will need to keep 5 significant figures (有效数字) in your calculations, so use these precise values for the constants:

$\hbar = 1.0546 \times 10^{-34}$ J · s ("h-bar" = Planck's constant/2π)

$N_A = 6.0221 \times 10^{23}$ molecules/mole (Avogadro's number)

$k_B = 1.3807 \times 10^{-23}$ J/K (Boltzmann's constant).

(a) What is one quantum of energy for one oscillator in copper? (Remember that, in the Einstein model, the effective spring stiffness is 4 times the true interatomic spring stiffness.)

$$E_{\text{one quantum}} = \hbar\sqrt{\frac{k_{s,\text{eff}}}{m_a}} = \hbar\sqrt{\frac{4\times k_{s,i}}{m_a}} = (1.0546\times10^{-34})\sqrt{\frac{4\times 28}{\left(\frac{0.064}{6.0221\times10^{23}}\right)}}$$

$$= \boxed{3.4236\times10^{-21}\ \text{J}}$$

$$(\approx 0.021\ \text{eV})$$

(b) The table below contains the number of ways to arrange a given number of quanta of energy in a particular block of copper. Fill in the blank spaces to complete the table, including calculating the temperature of the block. The energy E is measured relative to the ground state. Be sure to give the temperature to the nearest 0.1 K.

q	# ways	E, J $\times 10^{-20}$	S, J/K $\times 10^{-22}$	ΔE, J $\times 10^{-21}$	ΔS, J/K $\times 10^{-23}$	T, K
20	4.91 E26	6.847	8.4856			
				3.424	3.040	112.6
21	4.44 E27	7.190	8.7896			
				3.424	2.982	114.8
22	3.85 E28	7.532	9.0878			

$\Delta T = 2.2\,\mathrm{K}$

E: $q E_{\text{one quantum}}$ S: $k_B \ln \Omega$ T: $\dfrac{\Delta E}{\Delta S}$

(c) There are 60 atoms in this object (it is a copper nanoparticle - 纳米粒子). What is the specific heat capacity per atom? Compare your answer to the high-temperature limit of $C = 3k_B$ =4.2 ×10^{-23} J/K/atom.

From $q = 21.5$ to $q = 22.5$, $\Delta E = E_{\text{one quantum}}$

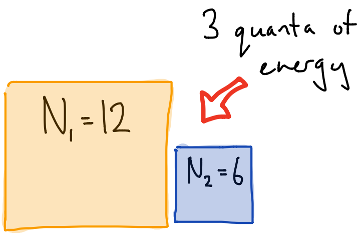
$$C = \frac{1}{N_{\text{Atoms}}}\left(\frac{\Delta E}{\Delta T}\right)$$

$$= \frac{1}{60}\left(\frac{3.42\times10^{-21}\ \mathrm{J/K}}{2.2\ \mathrm{K}}\right) = \boxed{2.6\times10^{-23}\ \mathrm{J/K\ per\ atom}}$$

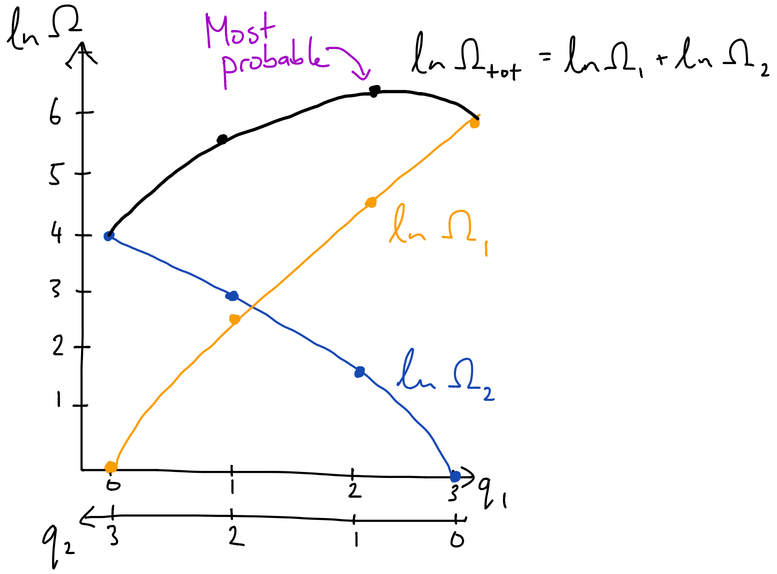
$$\simeq 0.6 \times 3k_B$$

(2) [M&I.7.P.46] A nanoparticle consisting of four iron atoms (object 1) initially has 1 quantum of energy. It is brought into contact with a nanoparticle consisting of two iron atoms (object 2), which initially has 2 quanta of energy. The mass of one mole of iron is 56 grams, and the interatomic spring stiffness is 45 N/m.

(a) Using the Einstein model of a solid, calculate and plot $\ln \Omega_1$ vs q_1 (the number of quanta in object 1), $\ln \Omega_2$ vs q_1, and $\ln \Omega_{\text{total}}$ vs q_1 (put all three plots on the same graph). Show your work and explain briefly.



q_1	q_2	Ω_1	Ω_2	$\ln \Omega_1$	$\ln \Omega_2$	$\ln \Omega_{tot}$
0	3	1	56	0	4.0	4.0
1	2	12	21	2.5	3.0	5.5
2	1	78	6	4.4	1.8	6.2
3	0	364	1	5.9	0	5.9



(b) Calculate the approximate temperature of the objects at equilibrium. State what assumptions or approximations you made.

The equilibrium state is the most probable one: $q_1 = 2$, $q_2 = 1$.

$$E_{\substack{\text{one} \\ \text{quantum}}} = \hbar \sqrt{\frac{4 \times k_{s,i}}{m_a}} = (1.05 \times 10^{-34}) \sqrt{\frac{4 \times 45}{\left(\frac{0.056}{6.02 \times 10^{23}}\right)}} = 4.6 \times 10^{-21} \text{ J.}$$

To find $T \simeq \frac{\Delta E}{\Delta S}$ around $q_1 = 2$, there are several different ways I can do it.

The simplest is just to take the difference between $q_1 = 1$ and $q_1 = 3$, for block 1:

$$T_{q_1=2} \simeq \frac{E_{q_1=3} - E_{q_1=1}}{S_{q_1=3} - S_{q_1=1}} = \frac{2 \times E_{\text{one quantum}}}{k_B(\ln \Omega_{q_1=3} - \ln \Omega_{q_1=1})}$$

$$= \frac{2 \times 4.6 \times 10^{-21}}{1.38 \times 10^{-23}(5.9 - 2.5)}$$

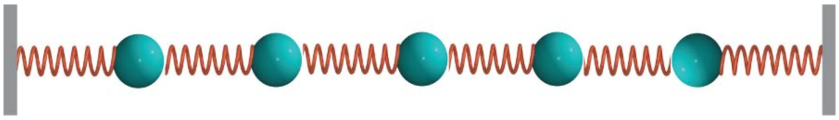
$$\simeq \boxed{200\ \text{K}}$$

I could also use $q_1 = 1 \to q_1 = 2$, to get $T(q_1 = 1.5)$

and $q_1 = 2 \to q_1 = 3$, to get $T(q_1 = 2.5)$,

then take the average of the two temperatures.

(3) [M&I.P.47] The figure below shows a one-dimensional (1D) row of 5 microscopic objects, each of mass 4×10^{-26} kg, connected by bonds that can be modeled as springs with stiffness 15 N/m. These objects can only move along the *x* axis.



(a) Using the Einstein model, calculate the approximate entropy of this system for total energy of 0, 1, 2, 3, 4, and 5 quanta. (Hint: this is 1D, not 3D, so one object corresponds to how many oscillators?)

In 1D, one object = one oscillator $\therefore N = 5$

q	Ω	$\ln \Omega$	$S \equiv k_B \ln \Omega$ $(\times 10^{-23}$ J/K$)$
0	1	0	0
1	$\frac{(1+5-1)!}{1!(5-1)!} = 5$	1.61	2.22
2	$\frac{(2+5-1)!}{2!(5-1)!} = 15$	2.71	3.74
3	$\frac{(3+5-1)!}{3!(5-1)!} = 35$	3.56	4.91
4	$\frac{(4+5-1)!}{4!(5-1)!} = 70$	4.25	5.87
5	$\frac{(5+5-1)!}{5!(5-1)!} = 126$	4.84	6.67

$\Delta S = 9.6 \times 10^{-24}$ J/K

$\Delta S = 8.0 \times 10^{-24}$ J/K

(b) Calculate the approximate temperature of the system when the total energy is 4 quanta.

$$E_{\text{one quantum}} = \hbar \sqrt{\frac{4 \times k_{s,i}}{m}} = (1.05 \times 10^{-34}) \sqrt{\frac{4 \times 15}{4 \times 10^{-26}}}$$

$$= 4.07 \times 10^{-21} \text{ J}$$

$q = 3 \rightarrow q = 4$ $\quad T \approx \frac{\Delta E}{\Delta S} = \frac{4.07 \times 10^{-21}}{9.6 \times 10^{-24}} = 424$ K $\quad (q = 3.5)$

average: $\boxed{467 \text{ K}}$ $\quad (q = 4)$

$q = 4 \rightarrow q = 5$ $\quad T \approx \frac{\Delta E}{\Delta S} = \frac{4.07 \times 10^{-21}}{8.0 \times 10^{-24}} = 509$ K $\quad (q = 4.5)$

specific heat per object

(c) Calculate the approximate ~~temperature~~ of the system when the total energy is 4 quanta.

$q = 3.5 \rightarrow q = 4.5$:

$$C = \frac{1}{N_{objects}}\left(\frac{\Delta E}{\Delta T}\right) = \frac{1}{5}\left(\frac{4.07\times10^{-21}\,\mathrm{J}}{[509-424]\,\mathrm{K}}\right)$$

$$= \boxed{9.6\times10^{-24}\ \mathrm{J/K}\ \text{per object}}$$

(d) If the temperature is raised very high, what is the approximate specific heat capacity per object? Compare with your result from part (c).

In 1D, $C = 1.\,k_B = \boxed{1.38\times10^{-23}\ \mathrm{J/K}}$ at high temperatures.

The result in part (c) is about $0.7\,k_B$.